

Parity-Renormalized First-Alias Packets: Exact Signs, Clock Phase, and the Neighboring-Shell Interface

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July 2026

Abstract

The first counterloop alias and the square-root parity leakage occur at the same weighted order, so their signs and normalization cannot be suppressed. We restore the Hardy scaling hidden in the earlier first-alias ledger and split the coefficient defect exactly into a raw-trace packet, a signed parity packet, a finite-radius radial correction, and the roots-of-unity alias impulse. At the even first alias the parity correction is positive, whereas the counterloop defect is subtracted from the total residual. The parity packet has a uniform moving-order expansion with an explicit quadratic remainder. On a fixed clock phase η , its leading ratio to the alias defect is $C_*C_M\lambda^\eta$, so the unique symbolic balance phase is $-\log(C_*C_M)/\log\lambda$. If $C_*C_M\lambda < 1$, this gives a strict scalar-only obstruction throughout the usual floor/ceil/nearest phase window $|\eta| \leq 1$. The archived decimal evaluation supports that hypothesis but is not an interval certificate. This conditional negative result does not exclude cancellation by the raw physical boundary packet and neighboring shell. We finish with a phase- and coordinate-preserving interface for that calculation. No joint full-trace replacement is proved.

1 Normalization and four packet types

Put

$$\lambda = 1.678573510428322\dots, \quad r_H = 0.85, \quad R = 1.4, \quad \beta = (r_H\sqrt{\lambda})^{-1}. \quad (1)$$

The number r_H is the chosen Hardy scaling radius, not a dynamical invariant. Let P_n be the deterministic physical flat trace and let K_σ be the row-normalized noisy Markov operator. The peripheral parity eigenvalue is written

$$\lambda_-(\sigma) = -(1 - \delta_\sigma), \quad \delta_\sigma = C_*\sqrt{\sigma} + o(\sqrt{\sigma}), \quad C_* = 0.105258535936908\dots \quad (2)$$

The decimal evaluates an archived positive integral; positivity and the square-root law, rather than the rounded decimal, are the analytic inputs.

We use explicit Hardy superscripts to avoid mixing two archived conventions:

$$c_{\sigma,n}^H = r_H^{-n} \{\text{Tr } K_\sigma^n - 1 - \lambda_-(\sigma)^n\}, \quad (3)$$

$$c_n^H = r_H^{-n} \{P_n - 1 - (-1)^n\}. \quad (4)$$

Define the raw-trace and parity packets

$$\mathcal{T}_{\sigma,n} = r_H^{-n} \{\text{Tr } K_\sigma^n - P_n\}, \quad (5)$$

$$\mathcal{P}_{\sigma,n} = r_H^{-n} \{(-1)^n - \lambda_-(\sigma)^n\} = (-1)^n r_H^{-n} \{1 - (1 - \delta_\sigma)^n\}. \quad (6)$$

Then, identically,

$$c_{\sigma,n}^H - c_n^H = \mathcal{T}_{\sigma,n} + \mathcal{P}_{\sigma,n}. \quad (7)$$

For $k \geq 2$, let M_k be the archived boundary-cycle multiplier and put

$$\rho_k = |M_k|^{-1/k}, \quad \beta_k = \frac{\sqrt{\rho_k}}{r_H} = \frac{|M_k|^{-1/(2k)}}{r_H}. \quad (8)$$

The Hardy-scaled finite counterloop is

$$s_{k,n} = \beta_k^n (2k \mathbf{1}_{2k|n} - 1 - (-1)^n), \quad \beta_k \longrightarrow \beta. \quad (9)$$

Rename the limiting pole moment and numerator anchor to avoid collisions with older uses of p_n and a_n :

$$p_n^{\text{pole}} = -2\mathbf{1}_{2|n}\beta^n, \quad a_n^{\text{num}} = c_n^H - p_n^{\text{pole}}. \quad (10)$$

2 Exact alias and parity algebra

Theorem 2.1 (Radial-plus-alias decomposition). *For every $k \geq 2$ and $n \geq 1$, the counterloop defect is exactly*

$$\mathcal{A}_{k,n} := s_{k,n} - p_n^{\text{pole}} = 2\mathbf{1}_{2|n}(\beta^n - \beta_k^n) + 2k\beta_k^n \mathbf{1}_{2k|n}. \quad (11)$$

The first term is finite-radius radial drift and the second is the exact roots-of-unity alias impulse. In particular, $\mathcal{A}_{k,n} = 0$ for every odd n , while

$$\mathcal{A}_{k,2k} = (2k - 2)\beta_k^{2k} + 2\beta^{2k} > 0. \quad (12)$$

Proof. Substitute (9) and (10). If n is odd, both $1 + (-1)^n$ and the two divisibility indicators vanish. If n is even, the non-alias part is $-2\beta_k^n + 2\beta^n$, and an order divisible by $2k$ receives the additional impulse $2k\beta_k^n$. Setting $n = 2k$ gives (12). $\square$

The sign ledger is now unambiguous:

order	parity packet $\mathcal{P}_{\sigma,n}$	counterloop defect $\mathcal{A}_{k,n}$	signs in the residual
odd	negative	0	$+\mathcal{P}$
even, $2k \nmid n$	positive	$2(\beta^n - \beta_k^n)$	$+\mathcal{P} - \mathcal{A}$
first alias $n = 2k$	positive	$(2k - 2)\beta_k^{2k} + 2\beta^{2k}$	$+\mathcal{P} - \mathcal{A}$

Only the parity sign is asserted in the middle row; the radial difference need not be assigned a sign without a finite- k radius inequality.

Theorem 2.2 (Parity-renormalized first-alias packet identity). *Let*

$$e_{\sigma,k,n} = c_{\sigma,n}^H - s_{k,n} - a_n^{\text{num}}. \quad (13)$$

Then at the first alias,

$$e_{\sigma,k,2k} = \mathcal{T}_{\sigma,2k} + \mathcal{P}_{\sigma,2k} - \mathcal{A}_{k,2k} \quad (14)$$

$$= r_H^{-2k} \{ \text{Tr } K_\sigma^{2k} - P_{2k} + 1 - \lambda_-(\sigma)^{2k} \} - \{ (2k - 2)\beta_k^{2k} + 2\beta^{2k} \}. \quad (15)$$

In particular, the scalar parity contribution in (15) is

$$r_H^{-2k} \{ 1 - \lambda_-(\sigma)^{2k} \} = r_H^{-2k} \{ 1 - (1 - \delta_\sigma)^{2k} \} > 0. \quad (16)$$

Proof. Using $a_n^{\text{num}} = c_n^H - p_n^{\text{pole}}$ in (13) gives

$$e_{\sigma,k,n} = (c_{\sigma,n}^H - c_n^H) - (s_{k,n} - p_n^{\text{pole}}).$$

Apply (7), theorem 2.1, and set $n = 2k$. Equation (16) follows because $0 < 1 - \delta_\sigma < 1$ for sufficiently small noise. $\square$

This theorem is an exact coefficient identity. It does not approximate $\mathcal{T}_{\sigma,2k}$ by a local Gaussian law and therefore does not assume the missing trace-observation theorem.

3 Uniform parity packet on the first-alias clock

Theorem 3.1 (Uniform scalar parity expansion). *For $0 \leq \delta < 1$ and every $n \geq 1$,*

$$|\mathcal{P}_{\sigma,n} - (-1)^n n \delta r_H^{-n}| \leq \frac{n(n-1)}{2} \delta^2 r_H^{-n}, \quad (17)$$

where the left side uses (6) with $\delta_\sigma = \delta$. Consequently, if

$$k_\sigma = \frac{\log(1/\sigma)}{2 \log \lambda} + O(1), \quad (18)$$

then

$$\mathcal{P}_{\sigma,2k_\sigma} = 2k_\sigma C_* \sqrt{\sigma} r_H^{-2k_\sigma} \{1 + o(1)\}. \quad (19)$$

Its weighted size and the weighted first-alias defect share the growth exponent

$$\chi = \frac{\log(\beta R)}{\log \lambda} = 0.463406944517002 \dots \quad (20)$$

Proof. Taylor's theorem for $f(\delta) = 1 - (1 - \delta)^n$ gives

$$f(\delta) = n\delta - \frac{n(n-1)}{2} (1 - \xi)^{n-2} \delta^2$$

for some $0 \leq \xi \leq \delta$, proving (17). On (18), $k_\sigma \delta_\sigma \rightarrow 0$, so the remainder is $o(k_\sigma \delta_\sigma r_H^{-2k_\sigma})$ and (19) follows.

The archived multiplier law is

$$\beta_k = \beta \exp \left[-\frac{\log C_M}{2k} + o(k^{-1}) \right], \quad C_M = 1.946342905200968 \dots \quad (21)$$

Hence

$$\frac{\mathcal{A}_{k,2k}}{2k} = \frac{\beta^{2k}}{C_M} \{1 + o(1)\}. \quad (22)$$

After multiplication by R^{2k} , this is a constant multiple of $(\beta R)^{2k}$. The weighted parity packet divided by $2k$ is $C_* \sqrt{\sigma} (R/r_H)^{2k} \{1 + o(1)\}$. Using $2k = \log(1/\sigma)/\log \lambda + O(1)$ gives the same exponent (20). $\square$

4 Clock phase and a scalar-only obstruction

The bounded integer phase must be retained:

$$\eta_\sigma = k_\sigma - \frac{\log(1/\sigma)}{2 \log \lambda}. \quad (23)$$

Theorem 4.1 (Phase law and unique symbolic scalar balance). *Along every natural-clock sequence with bounded η_σ ,*

$$\boxed{\frac{\mathcal{P}_{\sigma,2k_\sigma}}{\mathcal{A}_{k_\sigma,2k_\sigma}} = C_* C_M \lambda^{\eta_\sigma} \{1 + o(1)\}.} \quad (24)$$

Consequently, along a subsequence with $\eta_\sigma \rightarrow \eta$, the ratio tends to $C_* C_M \lambda^\eta$. The leading scalar packets balance at the unique symbolic phase

$$\eta_* = -\frac{\log(C_* C_M)}{\log \lambda}. \quad (25)$$

Proof. Divide (19) by (22) and use $\beta^{2k} = r_H^{-2k} \lambda^{-k}$:

$$\frac{\mathcal{P}_{\sigma,2k}}{\mathcal{A}_{k,2k}} = C_* C_M \sqrt{\sigma} \lambda^k \{1 + o(1)\}.$$

Equation (23) makes $\sqrt{\sigma} \lambda^k = \lambda^{\eta_\sigma}$, proving (24). Positivity of C_* , C_M , and $\lambda > 1$ makes the right side strictly increasing in η , so its value one has the unique solution (25). $\square$

Corollary 4.2 (Conditional canonical-phase scalar obstruction). *Put*

$$q_+ = C_* C_M \lambda. \quad (26)$$

Assume $q_+ < 1$, let $|\eta_\sigma| \leq 1$, and suppose $\mathcal{T}_{\sigma,2k} = o(\mathcal{A}_{k,2k})$. Then

$$|e_{\sigma,k,2k}| \geq \{1 - q_+ + o(1)\} \mathcal{A}_{k,2k}, \quad (27)$$

and this is not $o(kR^{-2k})$.

Proof. Equation (24) and $|\eta_\sigma| \leq 1$ give $\mathcal{P}_{\sigma,2k}/\mathcal{A}_{k,2k} \leq q_+ + o(1)$. Substitute into (14). Finally,

$$\frac{\mathcal{A}_{k,2k}}{kR^{-2k}} = \frac{2}{C_M} (\beta R)^{2k} \{1 + o(1)\} \rightarrow \infty$$

because $\beta R > 1$. $\square$

This is deliberately a conditional scalar-only negative result. The repository proves the symbolic corollary but does not contain a directed interval enclosure certifying $q_+ < 1$. The archived decimal values make the hypothesis numerically plausible; they are not used as a rigorous inequality. Even under that hypothesis, the result does not bound the actual raw trace packet and does not prove divergence of the combined physical residual.

The same phase controls the local clearance. If the archived boundary-cycle clearance is

$$\Delta_k^{\text{clr}} = C_b \lambda^{-2k} \{1 + o(1)\}, \quad C_b > 0, \quad (28)$$

then

$$d_{\sigma,k} := \frac{\Delta_k^{\text{clr}}}{\sigma} = C_b \lambda^{-2\eta_\sigma} \{1 + o(1)\}. \quad (29)$$

At scalar balance,

$$d_* = C_b (C_* C_M)^2. \quad (30)$$

Thus suppressing the clock phase would discard both the symbolic scalar balance ratio and the RH-322 entrance profile. The decimal evaluations of C_b and d_* are numerical diagnostics only.

5 Typed interface for the neighboring shell

RH-322–RH-324 retain the entrance clearance and the oriented affine frame

$$V \sim g_d, \quad U = -\alpha V - Z_1, \quad W = -\lambda U + Z_2, \quad (V, U, W) \text{ has orientation } (+, -, +), \quad (31)$$

where $\alpha = 2u_c$ and the output is compared after the center shift $\kappa_{\text{aff}} d = 2u_c \lambda d$. Their objects are forward probability laws of mass one. They do not yet define a signed trace observation, and RH-19 shows that the neighboring critical sibling cannot be hidden in a far Gaussian tail.

The exact handoff is therefore conditional but typed. A later theorem must construct, on the same clock, phase, coordinate frame, and Hardy normalization, a decomposition

$$\mathcal{T}_{\sigma,2k} = \mathcal{B}_{\sigma,k}(d, \eta; V, U, W, \mathfrak{t}) + \mathcal{S}_{\sigma,k}(d, \eta; \mathfrak{t}) + \mathcal{R}_{\sigma,k}. \quad (32)$$

Here $\mathfrak{t}$ is the still-missing trace observation, $\mathcal{B}$ is the physical boundary packet, $\mathcal{S}$ is the neighboring-shell packet, and $\mathcal{R}$ contains the second-leg, all-leg transport, and observation remainders. Combining (32) with theorem 2.2 gives the exact join

$$e_{\sigma,k,2k} = \mathcal{B}_{\sigma,k} + \mathcal{S}_{\sigma,k} + \mathcal{R}_{\sigma,k} + \mathcal{P}_{\sigma,2k} - \mathcal{A}_{k,2k}. \quad (33)$$

Consequently, if $\mathcal{R}_{\sigma,k} = o(kR^{-2k})$, the desired matching is equivalent to the signed condition

$$\boxed{\mathcal{B}_{\sigma,k} + \mathcal{S}_{\sigma,k} = \mathcal{A}_{k,2k} - \mathcal{P}_{\sigma,2k} + o(kR^{-2k})}. \quad (34)$$

Equation (34) is an interface, not a theorem that the packets on its left have been constructed.

6 Reproduction and claim boundary

The reproduction code checks the exact moment split, the parity signs, the uniform Taylor bound, the clock/clearance dictionary, and the scalar balance law. Ordinary floating-point substitution of the archived decimal constants gives the following diagnostic ledger:

η	$C_*C_M\lambda^\eta$	$C_b\lambda^{-2\eta}$
-1	0.122049588	1.298368749
0	0.204869205	0.460805149
1	0.343888020	0.163544745
3.060914914	1.000000000	0.019340633

These are evaluations of proved symbolic formulas, not fitted noisy traces or directed-rounding enclosures. In particular, the table does not certify $q_+ < 1$, $\eta_* > 1$, or the displayed clearance interval.

Remark 6.1 (Claim boundary). The exact identity combines only known algebraic packets. The phase law is symbolic, while the canonical-window obstruction is conditional on $q_+ < 1$ because the archived decimals are not interval certified. The paper does not identify the RH-322–RH-324 probability law with $\mathcal{T}_{\sigma,2k}$, control the second physical critical leg, prove all-leg phase transport, bound the weighted trace observation, include the neighboring shell, or establish (32). Hence no joint first-alias trace law, full-trace replacement, or actual divergence theorem is obtained. Gates A–E remain false/open. This paper constructs no Hilbert–Polya operator, identifies no Riemann zero, proves no von Mangoldt trace formula or zeta-divisor equality, and does not imply RH.

RH-327 should now compute the neighboring-shell coupling in the typed frame of (32); RH-328 may formulate the joint matching equation only after that shell and the trace observation are defined.